TEXORA · LECTURE SHEET

Trash and Foreign Matter

Trash is everything in the bale that is not spinnable fibre — leaf, seed coat, stem, dust and worse. It is weighed, graded and paid for before it is removed, so its measurement is both a commercial and a processing decision.

TE-3103 · QUALITY EVALUATION OF FIBRE AND YARN · TOPIC 7 · LEVEL 3 TERM 1

1. What counts as trash and foreign matter

- **Vegetable trash** — leaf and bract fragments, seed-coat pieces, stem and stalk particles, motes. The bulk of cotton trash; its size distribution matters (large particles are removed easily; fine "pepper" trash travels deep into the process).
- **Dust and micro-dust** — sub-millimetre particles, including broken fibre; a health concern in opening and card rooms and a contributor to yarn imperfections.
- **Foreign matter proper** — material that never grew on the plant: string, jute and polypropylene fragments from bale packaging, coloured fibres, oil, metal, stones. Small in mass, large in damage — a single length of polypropylene tape can ruin a dyed fabric lot because it does not take cotton dyes. Foreign-fibre contamination is hunted separately from trash percentage (contamination sorters, UV and camera systems, and bale-laydown discipline).

Trash content of raw cotton typically ranges from about 1 % (clean, hand-picked, roller-ginned cottons) to 6–8 % or more (machine-picked cottons). It directly reduces **mill yield**: every percent of trash is a percent bought, transported, and thrown away — plus the good fibre carried out with it during cleaning.

2. The Shirley Analyser — the reference method

The **Shirley Analyser** is a miniature opening-and-cleaning machine: a licker-in type roller working against a feed arrangement passes the specimen through a cleaning grid while an air current separates the heavy trash (which falls through the grid into a trash tray) from the lint (carried forward and re-collected). Procedure:

1. Weigh a conditioned specimen — classically 100 g (4 oz in the original instrument).
2. Pass it through the analyser; collect the trash from the tray and the cleaned lint.
3. Pass the lint through a **second time** (the standard two-pass procedure) to complete the separation; combine the trash fractions.

4. Weigh trash, cleaned lint, and account for the difference as **invisible loss** (dust and fly escaping with the air).

$$\text{Visible trash \%} = (\text{mass of trash collected} / \text{original specimen mass}) \times 100$$

$$\text{Total waste (analyser) \%} = \text{visible trash \%} + \text{invisible loss \%}$$

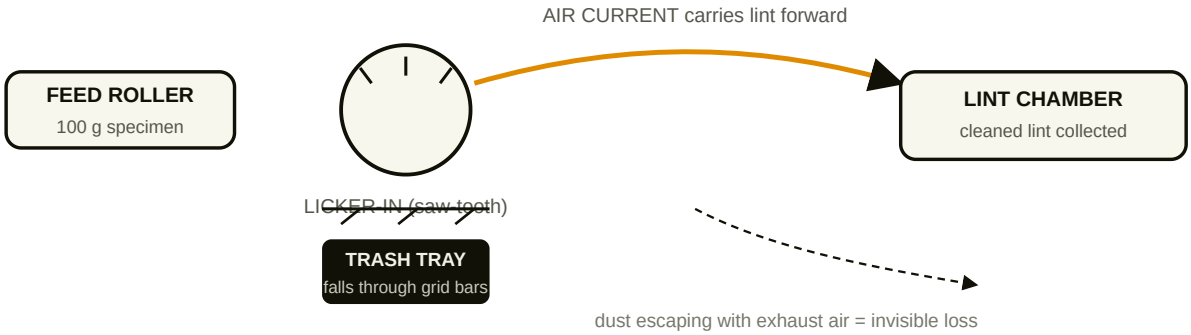


Figure 1 — Shirley Analyser flow. The licker-in opens the specimen over grid bars; gravity drops trash to the tray while the air stream carries lint to its chamber. The specimen is passed twice.

WORKED EXAMPLE — TRASH CONTENT AND TOTAL WASTE

A 100.0 g cotton specimen after two passes gives: cleaned lint 95.8 g, collected trash 3.4 g.
Invisible loss = 100.0 − 95.8 − 3.4 = 0.8 g.
Visible trash = 3.4 / 100.0 × 100 = **3.4 %**. Total analyser waste = 3.4 + 0.8 = **4.2 %**.
On a 170 kg bale of the same cotton the buyer should expect roughly 0.042 × 170 ≈ **7.1 kg** of analyser waste — before counting the further good-fibre losses of the mill's own cleaning line, which is why trash percentage is a pricing item and not a curiosity.

3. Other and modern methods

- **Classer's leaf grade.** Visual comparison of the sample against physical grade standards remains part of cotton classification (leaf grades 1–7 in the US system); it grades appearance, which correlates with, but is not identical to, weighed trash.
- **HVI trash measurement** (Topic 9) — a camera scans the specimen surface and reports trash particle count and % area covered; a surface measure, calibrated to correlate with gravimetric trash.
- **AFIS trash and dust counts** (Topic 9) — particles per gram by size class, distinguishing trash from dust and from neps.
- **MDTA / trash analysers** — rotor/opening-type instruments that clean a small sample and weigh the separated waste; used for sliver and roving waste checks in mills.

- **Gravimetric cleaning checks in the mill** — weighing feed and waste at each cleaning machine to establish cleaning efficiency and total blowroom/card waste, on the same mass-balance principle as the analyser.

4. Interpreting the result

- Trash % must be read with particle size: 3 % of large leaf cleans out almost fully at the blowroom; 3 % of fine pepper trash reaches the yarn.
- Cleaning is a trade-off: more aggressive cleaning removes more trash and more good fibre with it, and breaks fibre (lowering length, Topic 3) and creates neps (Topic 6). The optimum is the minimum total waste that meets the product's cleanliness.
- Seed-coat fragments deserve special attention: they survive cleaning, then show as dark specks and dye-resist spots — counted within trash but behaving like neps downstream.
- Report the method and the number of passes: a one-pass analyser figure understates the two-pass standard figure.

Glossary

Trash — non-fibre vegetable matter in the cotton: leaf, seed coat, stem, motes. **Foreign matter** — non-vegetable contamination: string, plastic, metal, oil, foreign fibres. **Visible trash** — trash physically collected and weighed in the analyser. **Invisible loss** — dust and fly escaping with the exhaust air, found by difference. **Shirley Analyser** — the standard laboratory opening/cleaning instrument for trash determination. **Leaf grade** — visual grade of leaf content against standard samples. **Cleaning efficiency** — percentage of incoming trash removed by a machine or line.

Key points

• Trash costs twice: it is bought by weight, then thrown away with some good fibre attached.
• Shirley Analyser, two passes on 100 g: report visible trash % and total waste % (with invisible loss) separately.
• Particle size decides cleanability; fine trash and seed-coat fragments are the stubborn fractions.
• HVI and AFIS give fast surface / particle-count correlates — the gravimetric analyser stays the reference.
• Foreign fibres (plastic, jute, coloured) are a separate contamination problem, hunted by sorters and discipline, not by trash %.

Practice

1. A 100.0 g specimen gives lint 96.5 g and trash 2.7 g. Find the visible trash % and the total waste %.
2. Two cottons both show 3.0 % trash; A's trash is coarse leaf, B's is fine pepper trash with seed-coat fragments. Which cleans more cheaply, and which threatens dyed fabric more?
3. Why does the analyser use two passes, and what error follows from using one?
4. A mill raises blowroom beating to cut yarn trash specks and finds yarn strength falling. Connect the three results.

Answer key

1. Invisible loss = 100.0 − 96.5 − 2.7 = 0.8 g. Visible trash = **2.7 %**; total waste = 2.7 + 0.8 = **3.5 %**.
2. A: large particles fall out at the first grids with little fibre loss — cheap cleaning. B's fine particles travel further, need harsher cleaning, and its seed-coat fragments survive to appear as specks that resist dye — B is the dyed-fabric risk.
3. One pass leaves part of the separable trash still entangled in the lint; the second pass completes the separation to the standard end-point. A one-pass figure systematically **understates** trash and is not comparable with standard results.
4. More beating removes more trash but also breaks fibres (shorter effective length, Topic 3) and throws good fibre into the waste. Yarn strength depends on fibre length and on the amount of sound fibre spun — both fall, so strength drops even though cleanliness improves. The correct fix is the minimum cleaning that meets specification, or a cleaner cotton.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib